

Pandas_Handson_Demography_Fermin

June 3, 2026

Guion

1- Hacer 2 chorradas de la intro 2- Abrir un fichero demográfico del 2000 3- Hacer 4 chorradas 4- Abrir el de 2007, ver que no es homogéneo, eliminar columnas, unificar nombres 5- dropna 6- Leer todos los años

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[ ]: import pandas as pd
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[ ]: import numpy as np
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[ ]: import matplotlib.pyplot as plt
      %matplotlib inline
```

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[ ]: data_file="pobmun/pobmun07.xls"
```

```
[ ]: column_names = ["PROVINCE", "TOWN", "TOTAL", "MEN", "WOMEN"]
      munpop=pd.read_excel(data_file,
      ↪skiprows=[0],parse_cols=[1,3,4,5,6],names=column_names)
```

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[ ]: munpop.columns
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```
[ ]: munpop
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[ ]: munpop[['PROVINCE','TOTAL']]
```

```
[ ]: province_pop = munpop.groupby('PROVINCE').TOTAL.sum()
      province_mun = munpop.groupby('PROVINCE').TOWN.count()

      province_pop.sort_values(inplace=True)
```

```
[ ]: province_pop[-10:].plot(kind='barh')
```

```
[ ]: munpop = munpop.dropna()
      province_pop = munpop.groupby('PROVINCE').TOTAL.sum()
      province_mun = munpop.groupby('PROVINCE').TOWN.count()

      province_pop.sort_values(inplace=True)
```

```
[ ]: province_pop[-10:].plot(kind='barh')

[ ]: province_mun.plot(kind='bar')

[ ]: province_pop['Girona']

[ ]: munpop.TOWN[munpop.WOMEN.argmax()]
munpop.TOTAL.max()

[ ]: munpop.describe()

[ ]: import os.path

[ ]: year_range = range(2000,2017)
pieces = []

column_names = ["PROVINCE", "TOWN", "TOTAL", "MEN", "WOMEN"]

for year in year_range:
    year_str = str(year)[-2:]
    data_file = 'pobmun/pobmun{}.xls'.format(year_str)
    if not os.path.isfile(data_file): data_file += "x"
    myframe=pd.read_excel(data_file,
↪skiprows=[0],parse_cols=[1,3,4,5,6],names=column_names)
    myframe['YEAR'] = year
    pieces.append(myframe)

[ ]: munpop = pd.concat(pieces,ignore_index=True)

[ ]: munpop = munpop.dropna()

[ ]: munpop.sum()

[ ]: #població PROVINCE
population = munpop.
↪pivot_table("TOTAL",index="YEAR",columns="PROVINCE",aggfunc=sum)

[ ]: import pandas as pd

[ ]: population.Barcelona.plot()
population.Madrid.plot()

[ ]: population.Navarra.plot()

[ ]:
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```
[ ]: #població TOWN
population = munpop.pivot_table("TOTAL",index="YEAR",columns="TOWN",aggfunc=sum)
men = munpop.pivot_table("MEN",index="YEAR",columns="TOWN",aggfunc=sum)
women = munpop.pivot_table("WOMEN",index="YEAR",columns="TOWN",aggfunc=sum)
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[ ]: men.Barcelona.plot(label='men')
women.Barcelona.plot(label='women')
plt.legend()
```

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[ ]:
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